Arnav Singhal

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Data Scientist specializing in deep learning for medical imaging, biomedical NLP, and ML systems deployment. Reduced reconstruction error by 34% in graduate thesis research on Fourier-augmented neural architectures, and built end-to-end pipelines spanning dynamic pricing, biomedical question-answering, and cardiovascular imaging. Skilled in Python, PyTorch, and cloud-based ML workflows (GCP), with a track record of translating research into deployed, containerized applications. Recently completed M.Sc. in Germany and now seeking Data Scientist roles in India.

Education

M.Sc. Data and Computer Science | Universität Heidelberg | Heidelberg, Germany *April 2023 - March 2026*

Major: Machine Learning and Deep Learning | Minor: Economics

Thesis: Evaluating Fourier Layers in DeepPET Architecture — A Comparative Study for Forward Modelling of the 2D Helmholtz Equation — mean relative error reduced from 7.12E-04 to 4.64E-04

- Designed an FFT-augmented U-Net (DeepPET) integrating spectral convolution layers with hybrid fusion and skip connections for 2D wave-field reconstruction. Best-performing architecture variant reduced mean relative error by 34% and validation loss by 62% versus the baseline model, validated via multimode evaluation and physics-aware metrics in TensorBoard.
- Evaluated model performance using training/validation curves and confusion matrices, applying analytical thinking and critical reasoning to interpret gradient-based visualizations and assess architectural impact on emotion detection.
- Applied critical reasoning to conduct comparative model studies, interpreting performance metrics to justify design choices and optimize predictive outcomes.

B.Tech. Computer Science and Engineering | Galgotias University | Greater Noida, India *2016 – 2020*

Experience

Data Analyst | Toluna Corporate | Gurugram, India *June 2021 – September 2022*

- Optimised data processing workflows on the Quantum survey analytics platform, completing every client deliverable ahead of committed timelines over one year.
- Led a team of 5 junior analysts, responsible for task allocation, reviews, and deadlines, sustaining 100% client satisfaction (CSAT) across all deliveries for the quarter.
- Communicated analytical insights to Procter & Gamble with 0 client-reported errors, successfully translating complex analytical findings into actionable recommendations that drove process improvements.

Jr. Data Analyst | E2E Research Services Pvt. Ltd | New Delhi, India *January 2020 – May 2021*

- Prepared, merged, and analyzed approximately 100 survey datasets per month in SPSS and Excel, implementing rigorous data quality checks to improve reporting accuracy and reliability.

Core Competencies

- **ML Focus Areas:** Medical Imaging, Scientific ML, Biomedical NLP, Explainable AI, Predictive Modelling, ML Deployment, Machine Learning
- **Machine Learning & Deep Learning:** Supervised & Unsupervised ML, Neural Networks, NLP, GenAI
- **Data Engineering & Processing:** Python, PyTorch, SQL
- **Cloud & MLOps:** Google Cloud Platform (GCP), Weights & Biases, Docker, Git
- **Model Deployment & Tools:** Streamlit, SimpleITK, CI/CD, Quantum (survey analytics platform)
- **ML Lifecycle Management:** Model training, experiment tracking, model monitoring
- **Libraries & Tools:** NumPy, Pandas, Matplotlib, SPSS, Excel
- **Research & LLMs:** Prompt Engineering, Transformers, BERT Models, Biomedical Text Mining, RAG, Statistics and Probability
- **Business & Governance:** Stakeholder Alignment, Data-Driven Decision-Making, Cross-Functional Collaboration
- **Languages:** English (Fluent), German (A2)

Key Projects

Simulated Dynamic Pricing Engine | *Python, ML, Streamlit, Docker | GitHub*

- Built a real-time pricing engine that predicts ride fares from contextual signals: demand, distance, weather, vehicle type, and area.
- Engineered fare features and a cost structure for 10k+ samples using base fare, per-km rate, and demand multiplier to generate realistic training targets.
- Evaluated and validated predictive models by comparing Linear Regression and Gradient Boosting with a Random Forest regressor; the Random Forest achieved RMSE 2.08, MAE 1.67, and R^2 0.49 on an 80/20 held-out split.
- Deployed the pricing model in an interactive Streamlit dashboard packaged in Docker, providing reproducible, instant price estimates for any input scenario.

Explainable Emotion Detection Using B-Cos Networks | *Deep Learning, Computer Vision, PyTorch | FER2013 | GitHub | Heidelberg University*

- Built and compared a standard CNN and a B-cos CNN for 7-class facial emotion recognition, employing hypothesis testing to statistically validate that the performance differences between model architectures were significant.
- Designed a controlled model-comparison experiment by training both models with identical settings (SGD with momentum, cross-entropy loss, batch size 64, learning-rate scheduling, horizontal-flip augmentation and normalization), isolating architecture as the difference.
- Evaluated and visualized model performance using training and validation curves, per-class confusion matrices, and gradient-based visualizations of the facial regions each model relied on.
- Found that the B-cos CNN matched the baseline's accuracy, with small gains on happiness and anger. Its maps focused on the eyes and mouth, while the standard CNN's heatmaps were diffuse. This shows interpretability without an accuracy cost, which matters in healthcare and human-computer interaction.

PubMedQA: Biomedical Literature Search System | *NLP, Biomedical AI, Streamlit, HuggingFace Transformers | GitHub | Heidelberg University*

- Built an end-to-end question-answering system over 9,999 PubMed abstracts (2013-2023) that lets users ask questions in natural language and get a direct answer instead of a list of articles.

- Indexed each article's title and abstract, keyed by PMID, as dense vectors using txtai with PubMedBERT embeddings, so retrieval matches on meaning rather than keywords.
- Designed a two-stage retrieve-then-extract pipeline. Semantic search finds the k closest articles, and a DistilBERT (SQuAD-distilled) QA model extracts the answer from their combined text.
- Deployed it as an interactive Streamlit app with a question box and a search button.
- Checked answers on a small test set of 3 questions using Jaccard similarity against reference answers (threshold > 0.25). Analysis of the failures pointed to weak context handling and irrelevant retrieval on broad queries such as "AI's role in healthcare".

Coronary Arteries Reconstruction from CTA Dataset | *Medical Imaging | Heidelberg University*

- Processed CTA volumes (.mhd) with SimpleITK and extracted artery centerlines using a CNN-based orientation tracker.
- Generated 3D vessel reconstructions (M×N×4) from the tracked centerlines.
- Achieved 2.71 mm mean centerline deviation, with 100% of points within 5 mm of ground truth.
- Explored stenosis simulation to model narrowed vessels for cardiovascular analysis.

Certifications

- **Student Innovation Lab – International Student Support Prototype** | Heidelberg University | July 2025
  - Completed Entrepreneurship I & II, Leadership & Team Dynamics, and Experience Design Thinking coursework; designed user journeys and prototypes using experience-driven product development.
  - Leveraged design thinking and analytical frameworks during the Student Innovation Lab to translate research findings into actionable recommendations for international student support prototypes.
- **Seminar: Evidentiality-Guided Generation for Knowledge-Intensive NLP Tasks** — Retrieval-Augmented Generation, Summer 2024
- **Hands-On Data Science** | Certificate | LinkedIn Learning | February 2022

Impact & Recognition

- Employee of the Month — Toluna Corporate (May 2022)
- Above and Beyond Award — Toluna Corporate (May 2022)